

Arjun Suryawanshi

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA

Expected May 2028

Bachelor of Science in Engineering for Computer Science

GPA: 3.96/4.00

Coursework: Data Structures & Algorithms, Operating Systems, Databases, Mathematics of Machine Learning, Cloud Computing, Systems Programming, Discrete Math, Probability, Linear Algebra, Multivariable Calculus

Activities: Engineering Student Activities Council (ESAC) (Senior Treasurer), Penn Debate Society (Competitor)

SKILLS

Languages: Java, Python, JavaScript/TypeScript, HTML/CSS, SQL (Postgres), C/C++

Frameworks/Tools: NumPy, Pandas, Matplotlib, Git/GitHub, AWS (DynamoDB, EC2, EMR, RDS), Docker, PyTorch, TensorFlow, Scikit-Learn, OpenCV, Pgmpy, Anthropic/Perplexity API, Apache Spark

WORK & RELEVANT EXPERIENCE

Regeneron Pharmaceuticals | Process Sciences and Technology Intern, Rensselaer, NY

May 2026 - Present

- Testing reuse of Hamilton-based LC plate resins, combines ML techniques utilizing AlphaFold-based molecular interaction prediction and ANCOVA analysis of product purity relative to AKTA manual column resins
- Building internal supply chain app to automate manual Veeva doc search when evaluating supply change notifications

CIS 1200 (Programming Languages & Techniques) | Head Teaching Assistant (TA), Philadelphia, PA

August 2025 – Present

- Selected as 1/4 Head TAs from ~35 TAs based on commitment to staff and course contributions, i.e. EdCamelBot, a Slack Bot designed to speed up internal Q&A Discussion and collect FAQ analytics via Ed API and GitHub Actions
- Redesigning the course to utilize Verbal Assessments, manage TA course and social events (new TA interviews/training, exam grading, staff discussions), teach weekly recitations, >450 online questions answered

New Bolton Center Endocrinology Lab at UPenn Vet | Computational Researcher, Kennett Square, PA

June 2024 – Sept. 2025

- Developed a TensorFlow-based model to classify horse behavior associated with disease, hand-labelled >3000 data samples and tested >50 models to reach approx. 82% accuracy in pose estimation
- Created a custom 24/7 4-camera suite with OpenCV-based triangulation and Random Forest classification

Baur Lab at the UPenn Perelman School of Medicine | Student Researcher, Philadelphia, PA

Sept. 2024 – May 2025

- Researched NAD precursor mitochondrial effects, found >30% survival rate difference; published in JCI, co-contributor on “NAD precursors prolong survival and improve cardiac phenotypes in a mouse model of Friedreich’s Ataxia”
- Updated Python LC/MALDI-MS pipelines to classify tissue; Identified 8 metabolites via metabolic biomarker analysis

EXTRACURRICULARS & KEY PROJECTS

Penn Electric Racing | Vehicle Dynamics Lead

September 2024 - Present

- Enhancing Quasi-Static Simulation Python Software – developed motor efficiency procedure to efficiently solve a non-invertible, circular torque-power limiting equation, precomputing a cached 2D LUT to increase accuracy by ~38% in O(1) amortized time; integrated a new aerodynamics map to improve downforce calculations by ~12%
- Designed Anti-Roll Bars for 2025 car’s suspension – leading to wider and 30% more accurate ARB tuning range

FedCast | Bridgewater Associates Immersion: AI for Innovation Hackathon (1 of 24 students selected)

April 2026

- Built central bank (Fed/ECB) worldview inference pipeline to convert ~10,200 unstructured meeting transcripts over a 13-year window into 5 economic dimensions with predicted importance and projection vectors. Modeled via pgmpy Bayesian-network belief model with multi-step reasoning via LangChain/LangGraph and Perplexity API context retrieval
- Backtested over ~2300 belief snapshots, achieving 90% hike/hold/cut classification accuracy
- Shipped Dockerized FastAPI service with cached forecasts, job-based progress polling, and an interactive JS dashboard

Vanstagram

November – December 2025

- Built Vanstagram, a scalable social media platform with a dynamic frontend using JavaScript and HTML/CSS, live chat system using Socket.io, and friend to friend graph visualization using Vis.js, hosted on EC2 for secure cloud deployment
- Developed backend with 11 AWS DynamoDB tables for persistent and low-latency posts and comments, S3 for user images, and Java Apache Spark & EMR for personalized news feed via an adsorption-based recommendation engine

AWARDS

- **Regeneron Science Talent Search (STS) Award (2024):** Earned for developing the ArmLev, a biomedical arm tremor stabilization device, activating within 0.3 seconds with >90% reliability over a 12Hz range of tremors
- **Math Olympiads (2022-2024):** 3x American Invitational Mathematics Examination (AIME) Qualifier